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The extract and load data pipeline pattern

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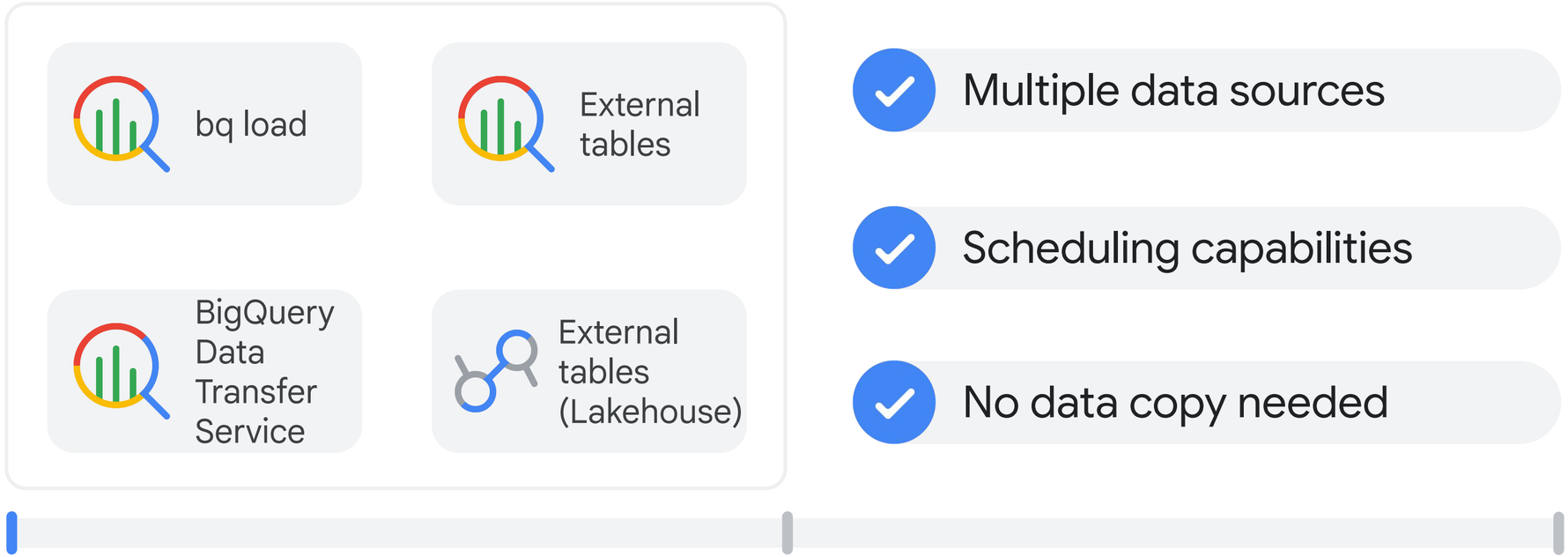
The extract and load data pipeline pattern

01	Extract and load architecture
02	The bq command line tool
03	BigQuery Data Transfer Service
04	Lakehouse for Apache Iceberg



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Extract and load ingests data into BigQuery without the need for transformation

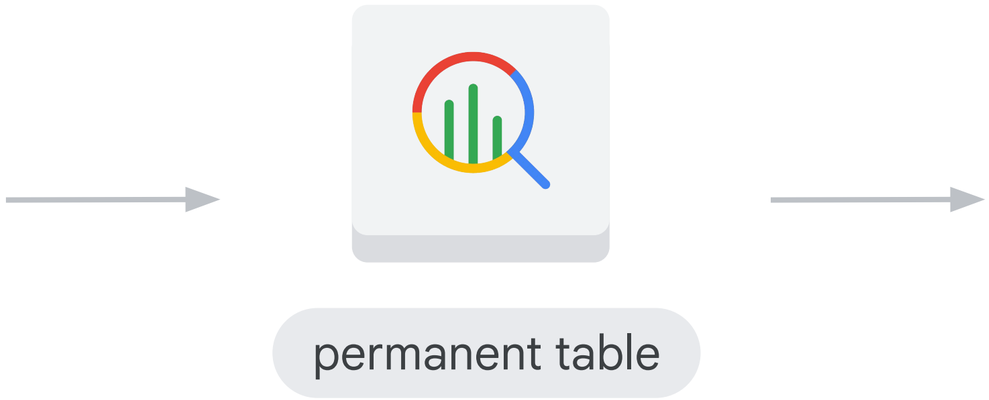


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BigQuery supports importing from and exporting to multiple formats

Loading data

- Avro
- Parquet
- ORC
- CSV
- JSON
- Firestore export



Query results

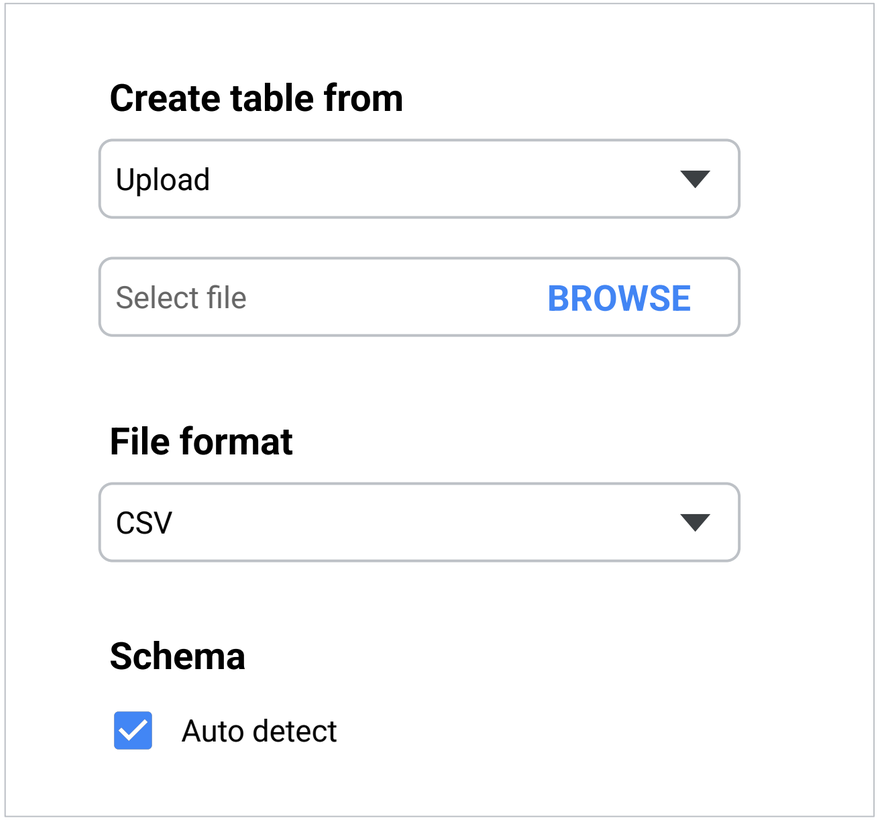
- CSV
- JSON
- Google Sheets

Table export

- CSV
- JSON
- Avro
- Parquet

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Load data into BigQuery by using the UI or the `LOAD DATA` statement




```
# create a new table by loading data from
# CSV files using schema auto-detect
LOAD DATA INTO dataset_name.table_name
 FROM FILES(
   format='CSV',
   uris = ['gs://mybucket/*.csv']
 )
```

- Use `LOAD DATA INTO` to append to an existing table.
- Use `LOAD DATA OVERWRITE` to overwrite the existing table.

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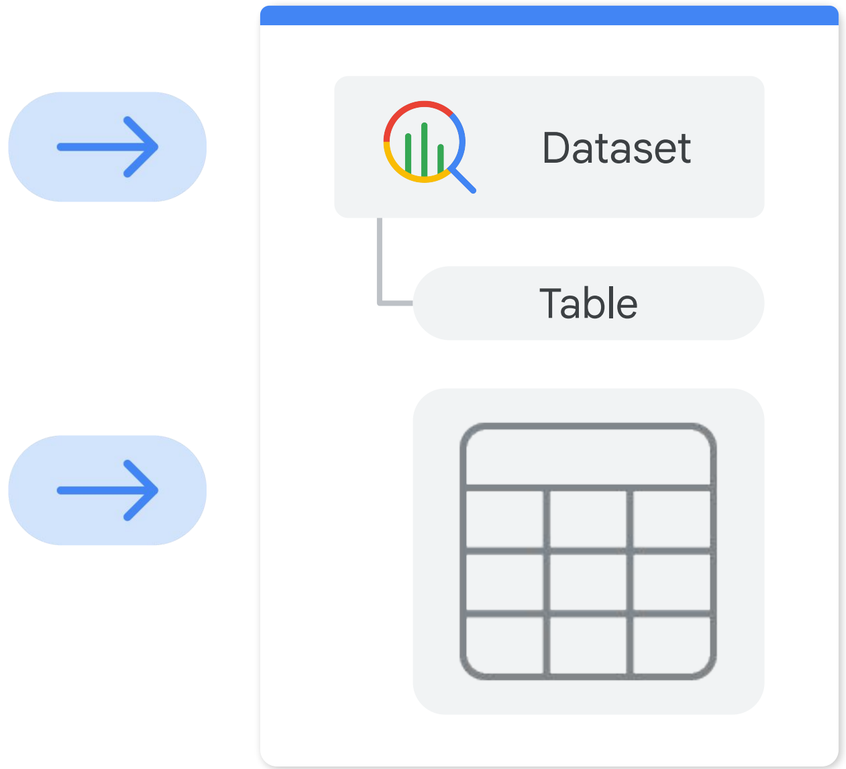


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Interact with BigQuery by using the bq command

```
>   bq mk --location=US -dataset dataset-name
```

```
>   bq load \
    --source_format=CSV \
    --skip_leading_rows=2
    dataset-name.table_name \
    "gs://mybucket/00/*.csv","gs://mybucket/01/*.csv" \
    ./table_schema.json
```



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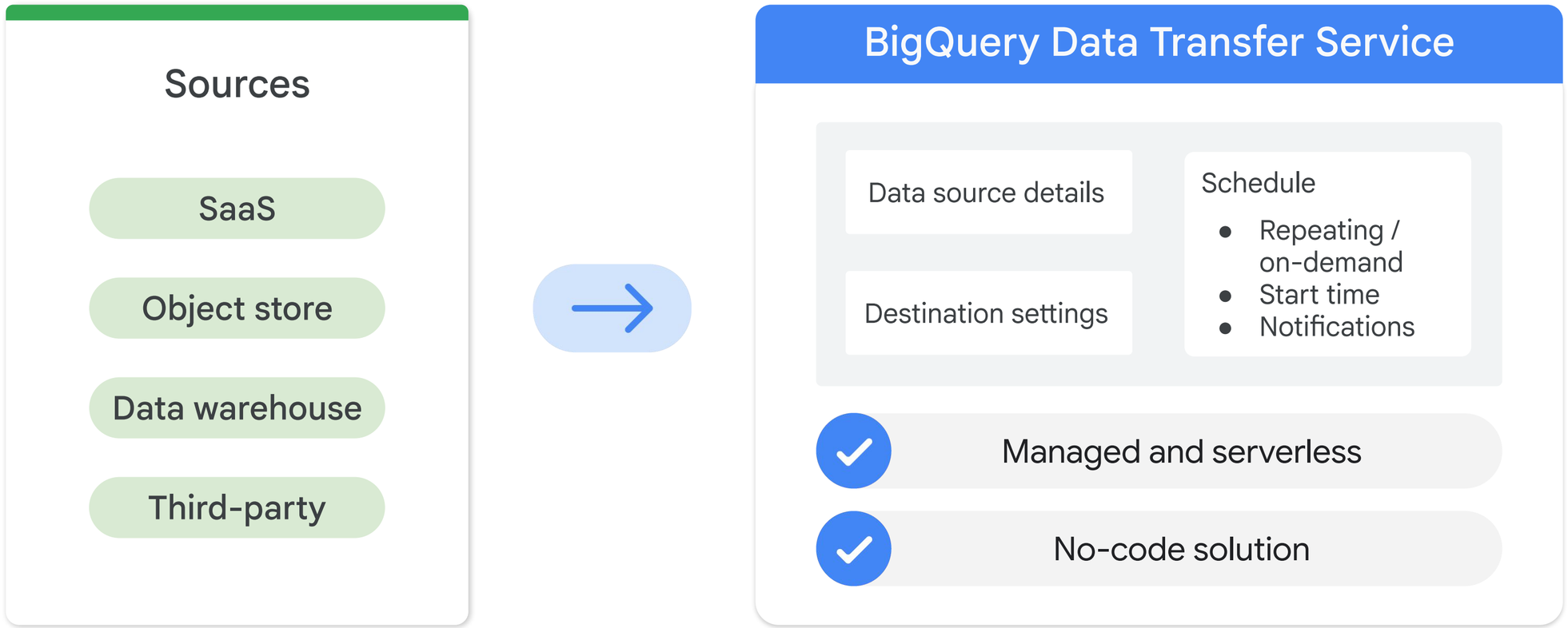
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Use the BigQuery Data Transfer Service to load data from other structured data sources



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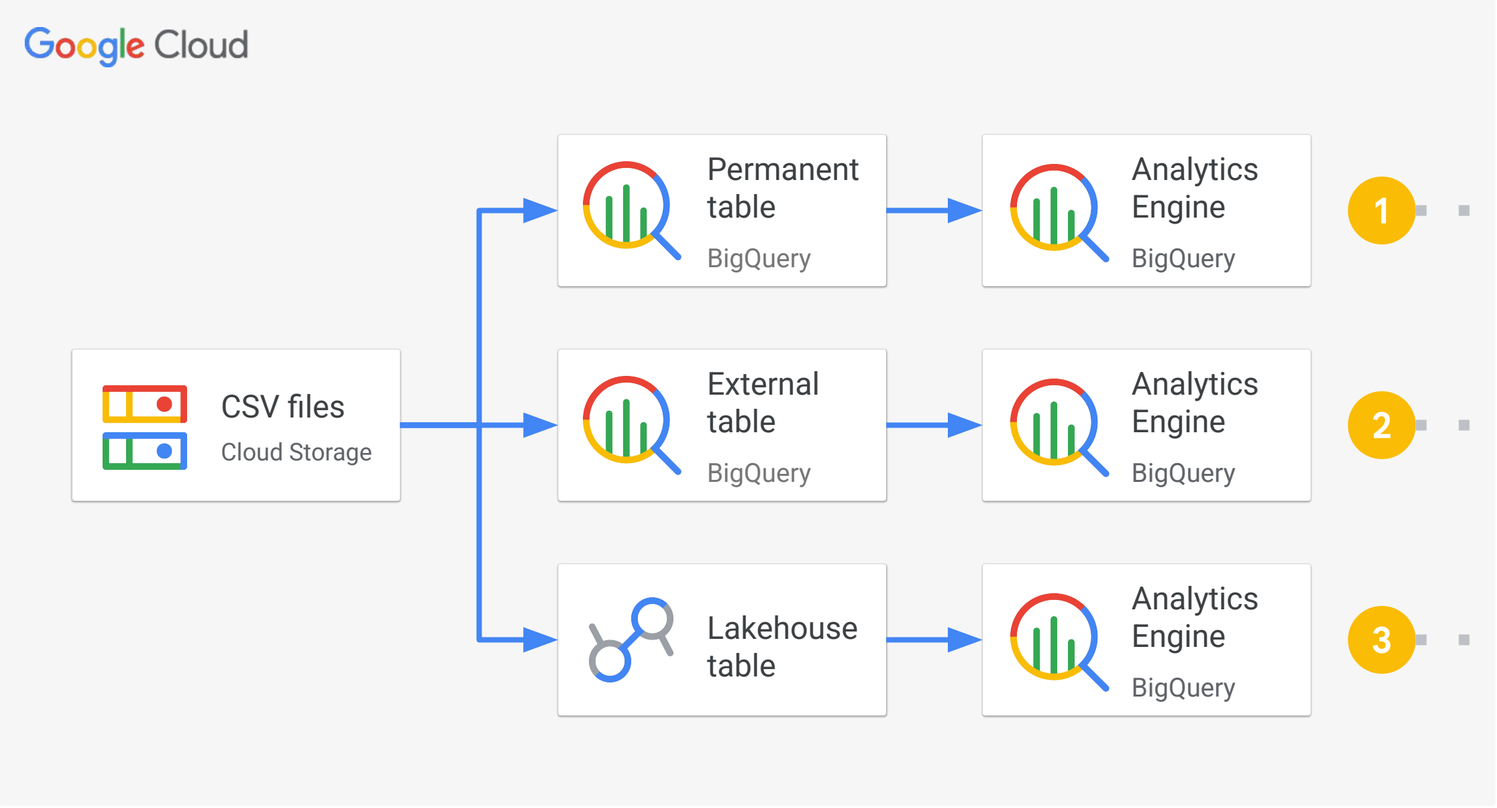
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Three ways of analyzing structured data in BigQuery



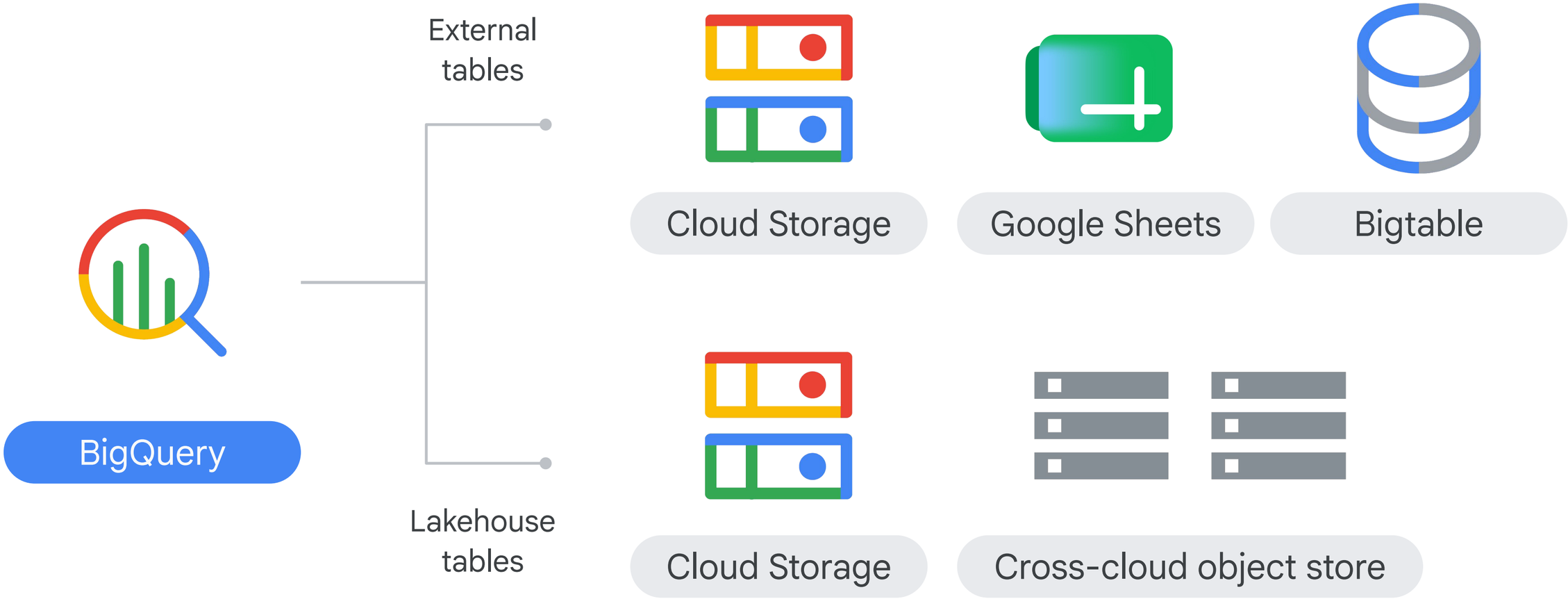
Load data into BigQuery storage, obtain high performance, but data movement is required.

Skip BigQuery storage, but receive low performance, with no data movement required.

Skip BigQuery storage, and receive high performance, with no data movement required.

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Your data isn't stored in BigQuery?
You can still query it!



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Compare external and Lakehouse tables

	External tables	Lakehouse tables
Query Engine	BigQuery	BigQuery, others using BigQuery connector
Data types	• Parquet, Avro, JSON, ORC, CSV • Datastore export, Firestore export • Google Sheets	• Parquet, Avro, JSON, ORC, CSV • Apache Iceberg, Delta, Hudi • Hive partitioned • Any unstructured (via object tables)
Locations	• Cloud Storage • Bigtable • Google Drive	• Cloud Storage • Amazon S3 • Azure Blob Storage
Access pattern	Direct link to the source. The user needs to have access to the source	External connection with access delegation. The service account has access to the source.
Security	None	• Column-level security, data masking • Row-level security

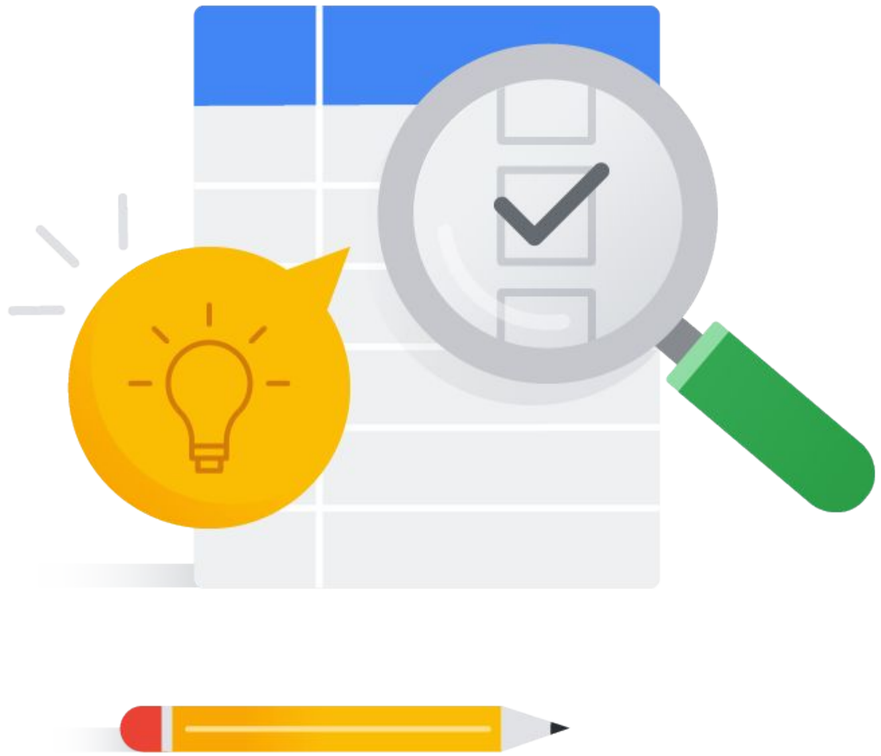
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Lab: Lakehouse: Qwik Start

45 min

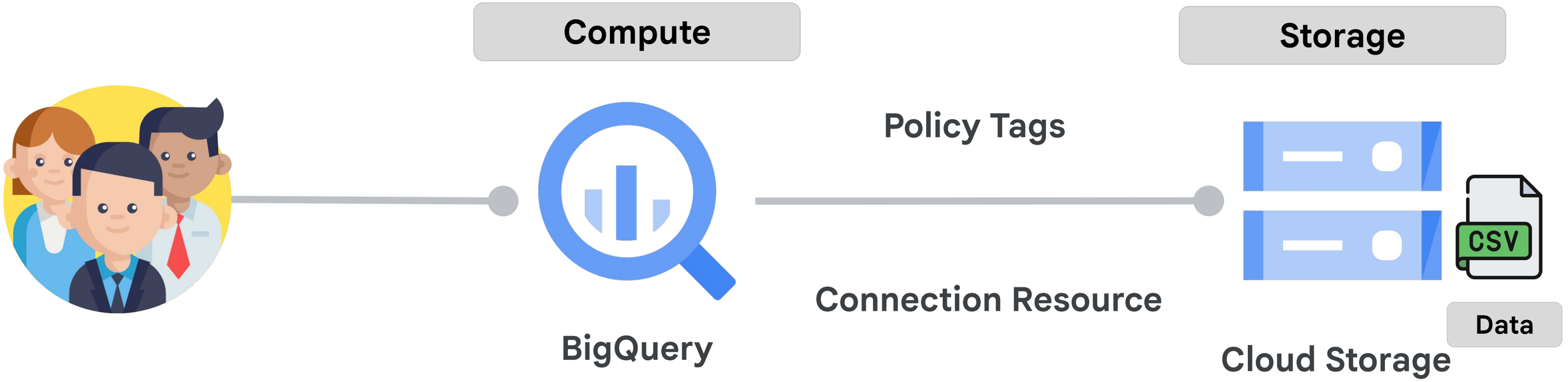
Learning objectives

- Create and view a connection resource.
- Set up access to a Cloud Storage data lake.
- Create a Lakehouse table.
- Query a Lakehouse table through BigQuery.
- Set up access control policies.
- Upgrade an external table to be a Lakehouse table.



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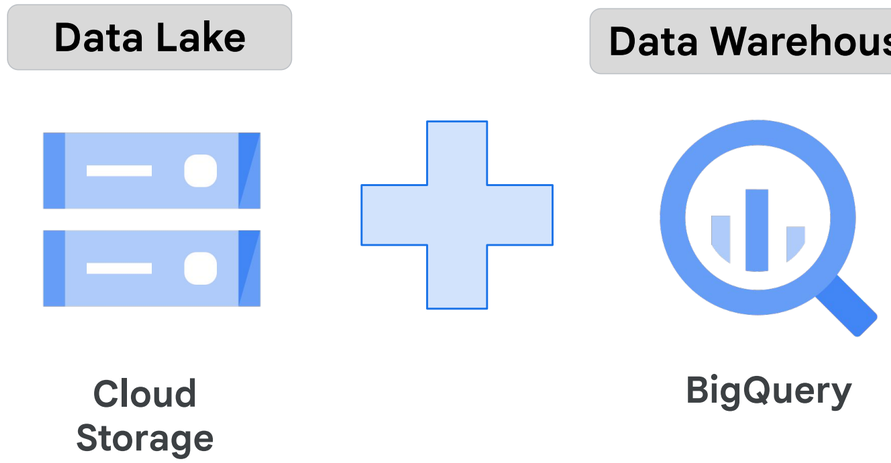
Lab: Lakehouse: Qwik Start



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Key Highlight: Lakehouse



1	**Low Storage Cost**: ค่าใช้จ่ายถูกขึ้น เนื่องจาก Data ถูกเก็บอยู่ใน Cloud Storage ไม่ใช่ BigQuery
2	**Zero ETL**: ไม่ต้องเสียเวลาทำ ETL ย้าย Data ไปมาระหว่าง Cloud Storage และ BigQuery
3	**Data Masking:** รองรับการทำ Column & Row level security ซึ่ง Cloud Storage ปกติไม่รองรับ
4	**Avoid Vendor Lock-in:** ไม่ต้องกังวลว่าจะโดน Lock-in ให้ใช้แค่ BigQuery เนื่องจากเราสามารถเลือก Open Data Format อย่าง Parquet, CSV, JSON หรือ Avro ได้


Google Cloud